AZURE MASTER ARCHITECT PROGRAM · JUNE 2026

Unified Digital Citizen Services Platform.

One front door for 2.1 M Nordic citizens across DK · SE · NO

ARCHITECTURE, AI & AGENTIC SUBMISSION

What this is — and what it isn't.

🎯 A target vision
The production-target architecture for a federated Nordic platform — end to end.

⚙️ A working core
Deployed on a real Azure tenant with one command — the live demo runs on it.

📐 Some blueprint
A few bricks are designed and registered, not yet switched on — each with a roadmap gate.

Every claim in this deck carries one honesty label:

- Live — on the tenant today
- Implemented — merged & smoke-tested
- Scripted — installer phase, idempotent
- Blueprint — designed, not yet live
- Roadmap — external dependency

UDCSP is a **production accelerator**: a target vision, a working core you can run today, and a named path for the rest.

The problem.

2.1 M

citizens across DK SE NO

47

separate legacy portals

28 d

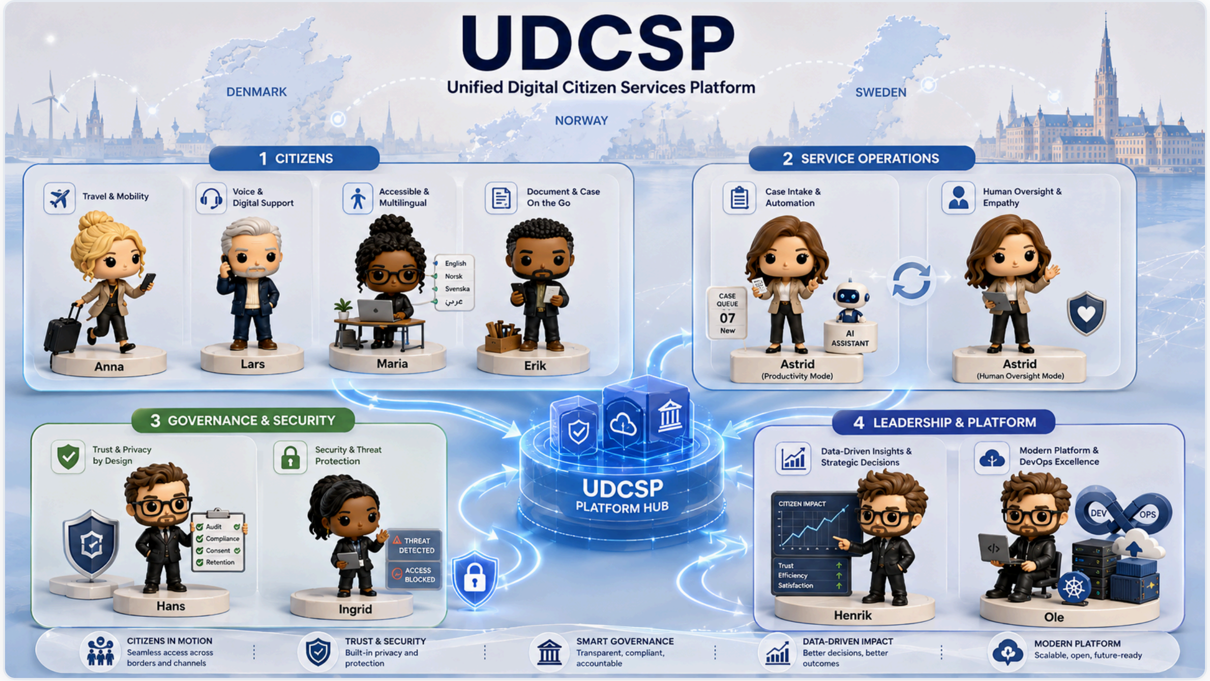
average time for a residency case

A citizen who moves from Copenhagen to Stockholm proves their identity again, re-sends the same papers, waits **28 days** for a decision, and uses a portal that may not speak their language — or may not be accessible at all.

UDCSP unifies the **identity** and the **experience**, then connects to the national systems to **share and check** information — never to replace them.

GDPR · EU AI Act · eIDAS 2.0 · NIS2 · WCAG 2.1 AA · ePrivacy

Who we build for.



CITIZENS

Anna · Lars · Maria · Erik

Anna moves country. Lars is blind and calls in. Maria uses a screen reader. Erik uploads from his phone.

OPERATIONS

Astrid — caseworker

The AI helps her work faster. She always makes the final decision.

GOVERNANCE

Hans · Ingrid

Hans protects privacy and consent. Ingrid watches for security threats.

LEADERSHIP

Henrik · Ole

Henrik needs clear data to decide. Ole runs the platform with one command.

The insight: unify the identity, connect the systems.

We did **not** rebuild the national portals. We unified the one thing that was missing — a single **identity** and a single **experience** — and connected it to the systems that already hold the records.

The citizen signs in once with their national **eID**. From there the platform **shares and checks** information with each authority: it pre-fills the right national form, submits it, and follows the status live.

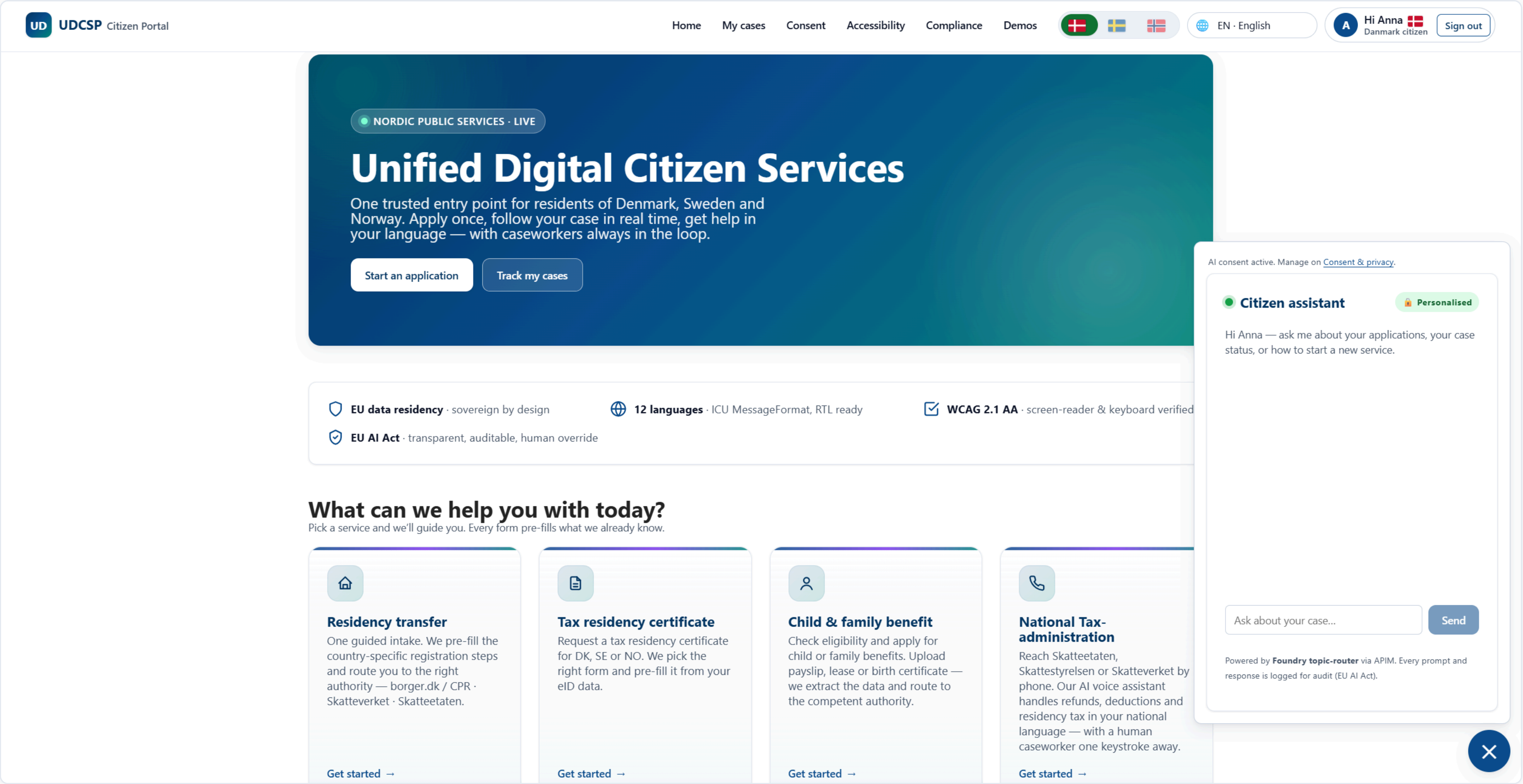
borger.dk, Skatteverket, NAV, Altinn and UDI stay the system of record and the decision-maker. We **connect** to them — we never replace them.

WHY IDENTITY-FIRST, NOT ANOTHER PORTAL

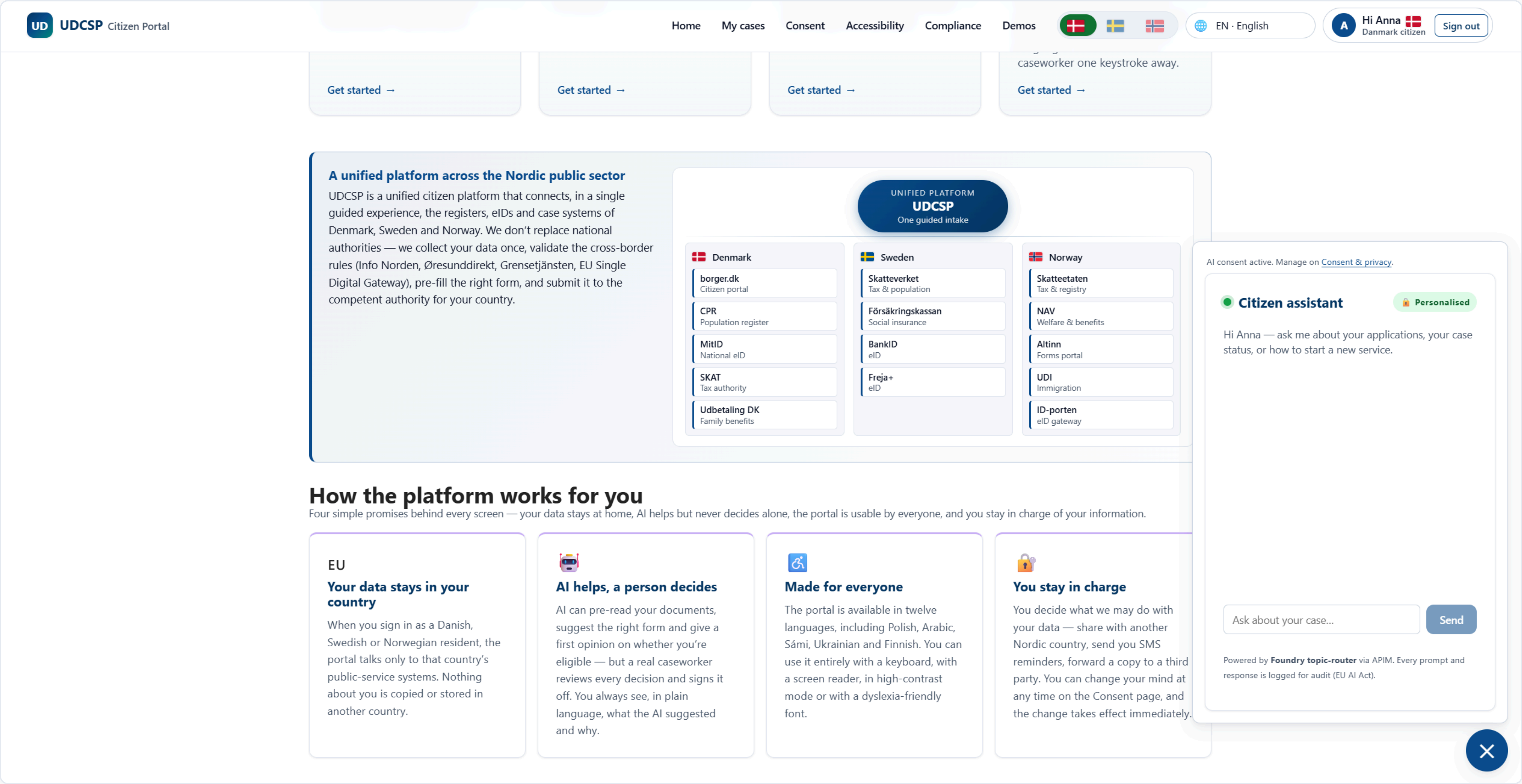
- One identity, one design, one accessibility standard
- Information is **shared and verified**, not re-collected
- The **same code** on web, mobile and phone
- A new service is a new card on the same shelf — not a new website
- The authorities keep their data; we keep the experience

> Sign in once. Share and check — don't re-collect. Follow the case live.

One front door — what the citizen lands on.



One identity, connected to every national system.

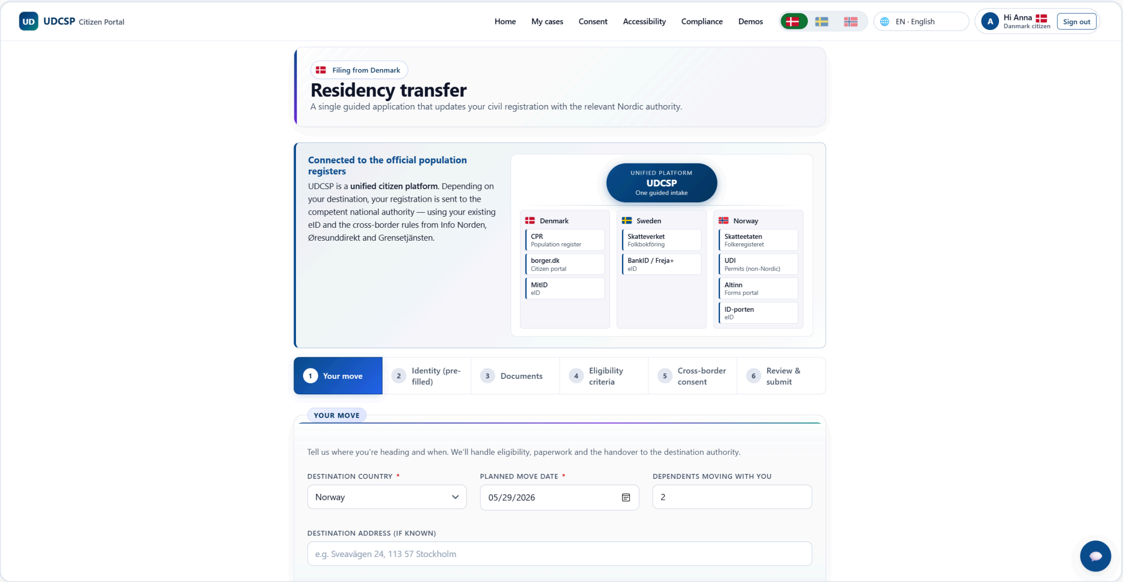


How the front door routes.

One guided intake. The portal finds **which authority owns the request** and sends it there — using each citizen's national eID and the public cross-border rules.

- **Denmark** — borger.dk · MitID
- **Sweden** — Skatteverket · BankID / Freja+
- **Norway** — UDI · Altinn · ID-porten

The rules come from Info Norden, Øresunddirekt and the EU Single Digital Gateway — public policy, not logic we invented.



What we ship.

CHANNEL

🌐 **Web · Mobile · Voice**

One web app · 12 languages · accessibility-tested · one free phone number per country

AI BRAIN

🧠 **One brain · 7 agents**

One orchestrator sends work to six specialists on Azure AI Foundry

OUTCOME

📉 **28 days → 4**

Cross-border residency in four days · 47 portals become one · 12 languages

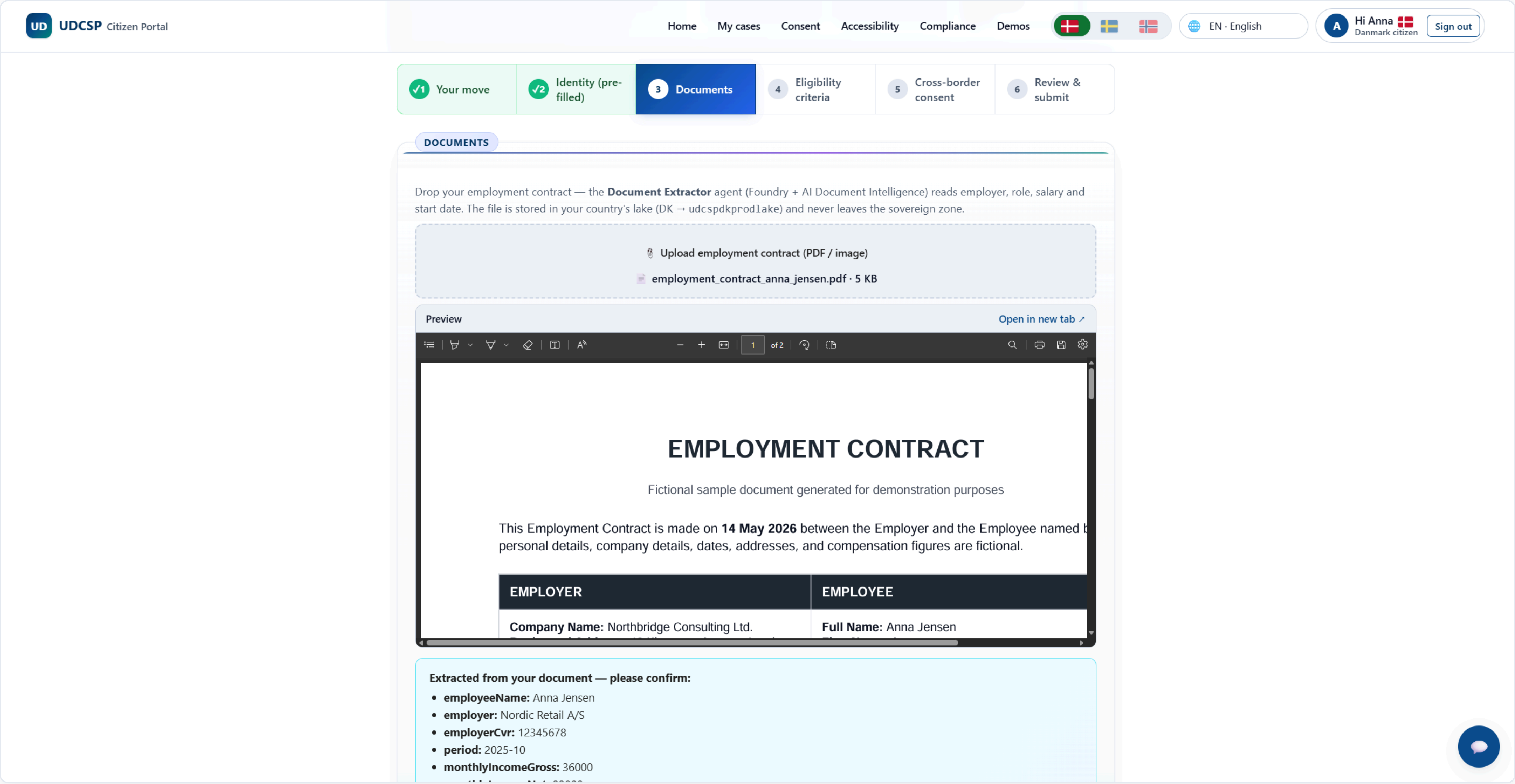
SOVEREIGN

🛡️ **3 zones**

One AI hub and one data zone per country · no data crosses a border uninvited

UDCSP is a **unified citizen platform**, not a replacement. The national authorities still make the decision.

Sign in once — the rest is pre-filled.



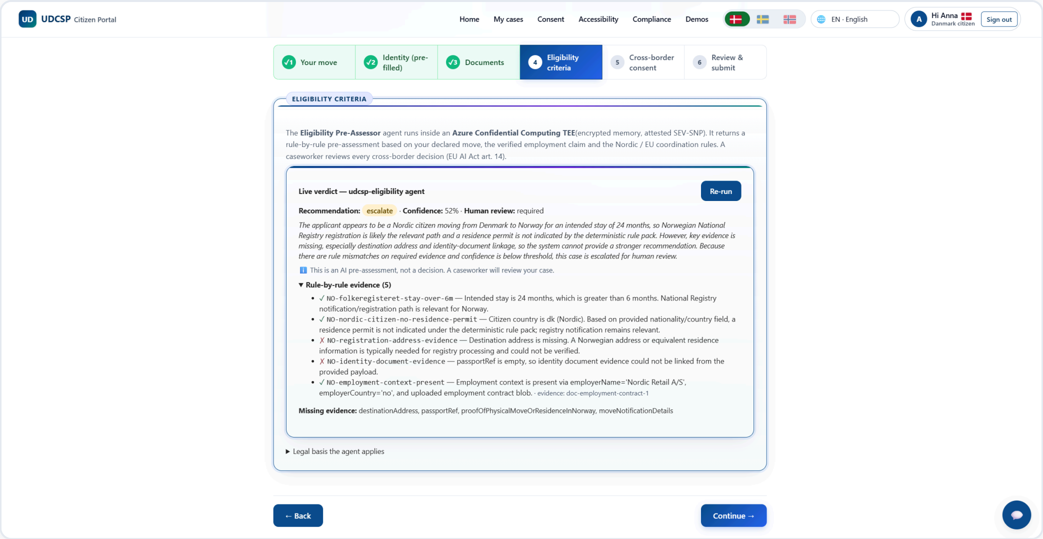
The platform you actually see.

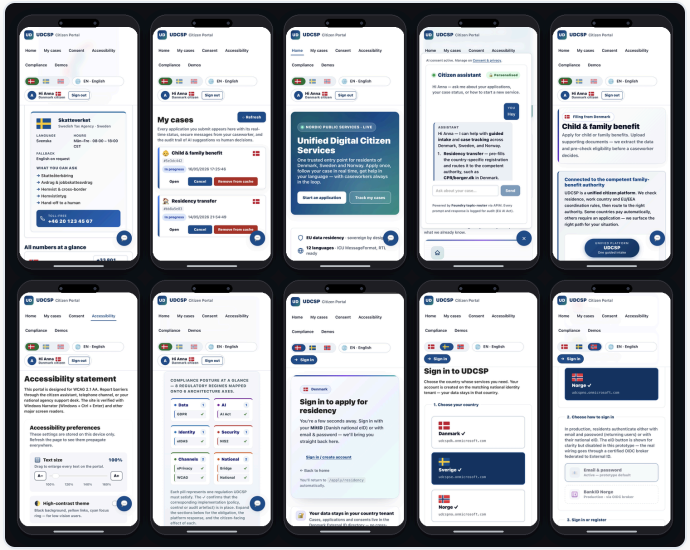
One web app · 12 languages · 3 channels · always accessible.

The same React and TypeScript code runs on phone, tablet and laptop, and adapts to each screen.

The chat assistant stays in the corner. The accessibility menu offers slow speech, high contrast and reduced motion.

On every *Apply* page, the citizen sees the AI result — confidence, evidence, missing documents — **before** they consent.



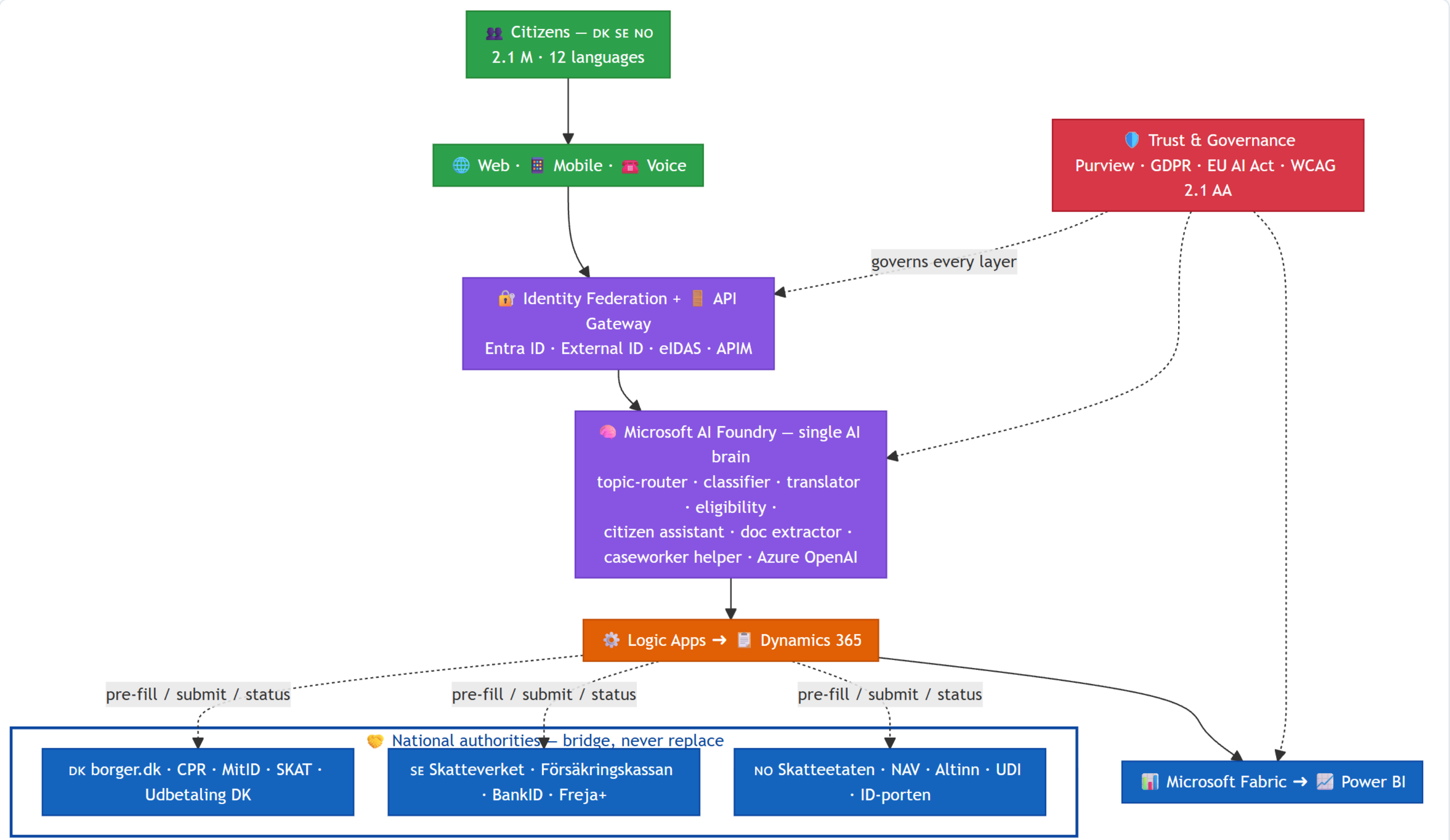


Architecture.

How we built it — three sovereign zones, one private path from the portal to the case

01

High-level architecture.



One country, one hub — three times.

A HUB PER COUNTRY

Each country has its own Azure region, private network, identity system, and **AI hub** on Azure AI Foundry.

- **Denmark** — North Europe region
- **Sweden** — Sweden Central region
- **Norway** — Norway East region

No AI call ever crosses a border.

THE RIGHT MODEL FOR EACH JOB

- **Real-time speech** model — the voice channel
- **Strong reasoning** model — the hard decisions
- **Fast** model — simple routing and sorting

The model name comes from one parameter file, so a region change is a one-line edit.

> **One honest exception.** Real-time speech is not available yet in Norway, so Norwegian calls use the Sweden hub, inside the EU Data Boundary. The audio still stays in Norway. When the model arrives, one parameter brings it home.

Privatising every zone — from the portal to the case.

One public door per country. Everything behind it is private — from the citizen's first click to the caseworker's queue.

01 **One public front door**

Azure Front Door and a web firewall. Nothing else is public.

02 **A private API gateway**

Azure API Management checks the token — the only way in.

03 **Backends on private endpoints**

Workflows, agents, storage, database, vault: off the internet. Private DNS per country.

04 **The case lands in Dynamics 365**

One environment per country, in-region. Writes go through workflows, reads through the gateway.

05 **Analytics copies only numbers**

Dynamics sends metrics to Microsoft Fabric. The raw rows stay in the country.

THE ONLY OTHER PUBLIC ADDRESS

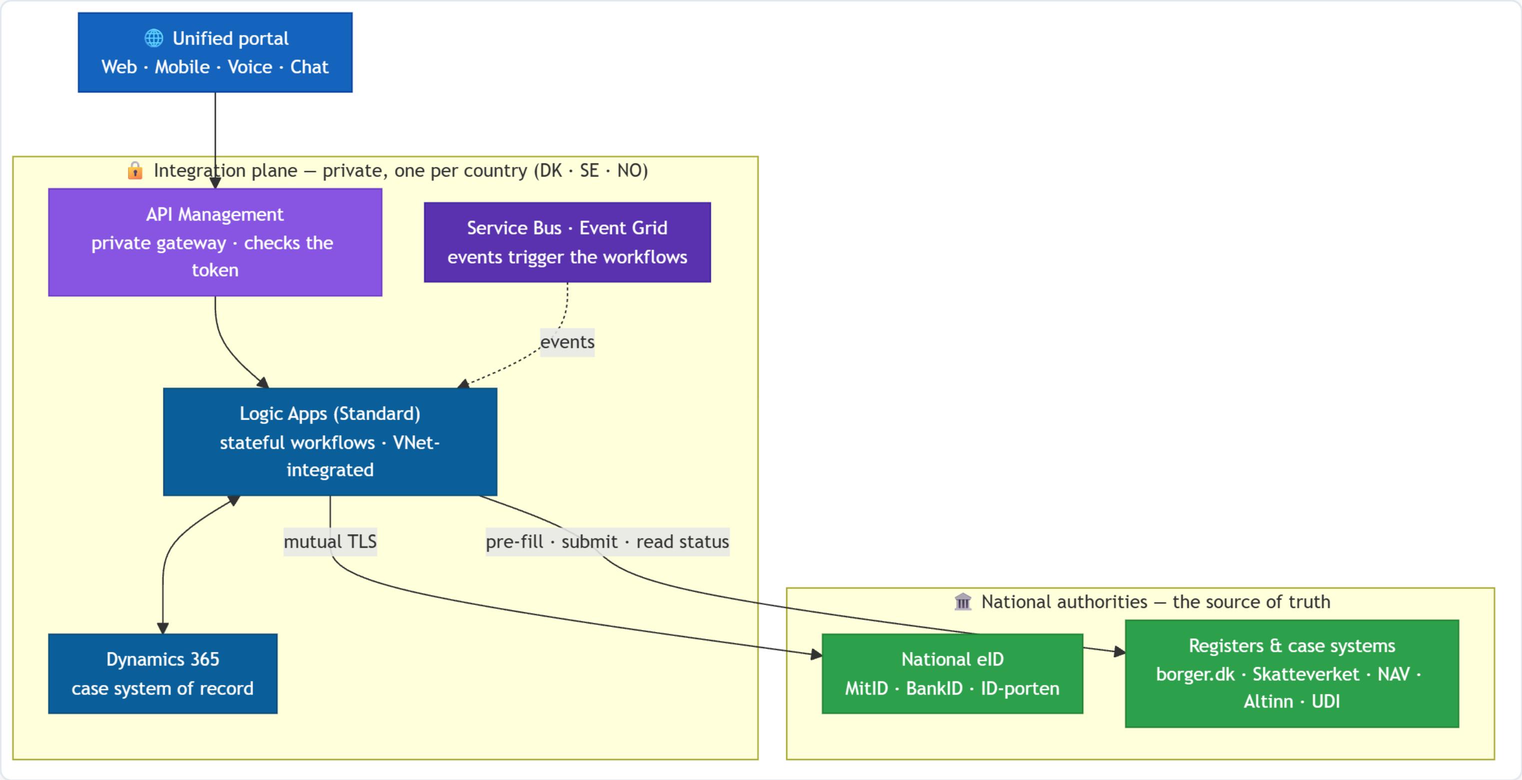
Azure Bastion — the single admin door per country. All outbound traffic goes through Azure Firewall, with allow-lists and TLS inspection.

ENCRYPTED EVERYWHERE

- Customer-managed keys, one set per country
- Two-way TLS to the national authorities
- Field-level encryption for national ID numbers

The integration bridge.

One private integration plane per country. The portal never calls a ministry directly — Logic Apps do, on its behalf.

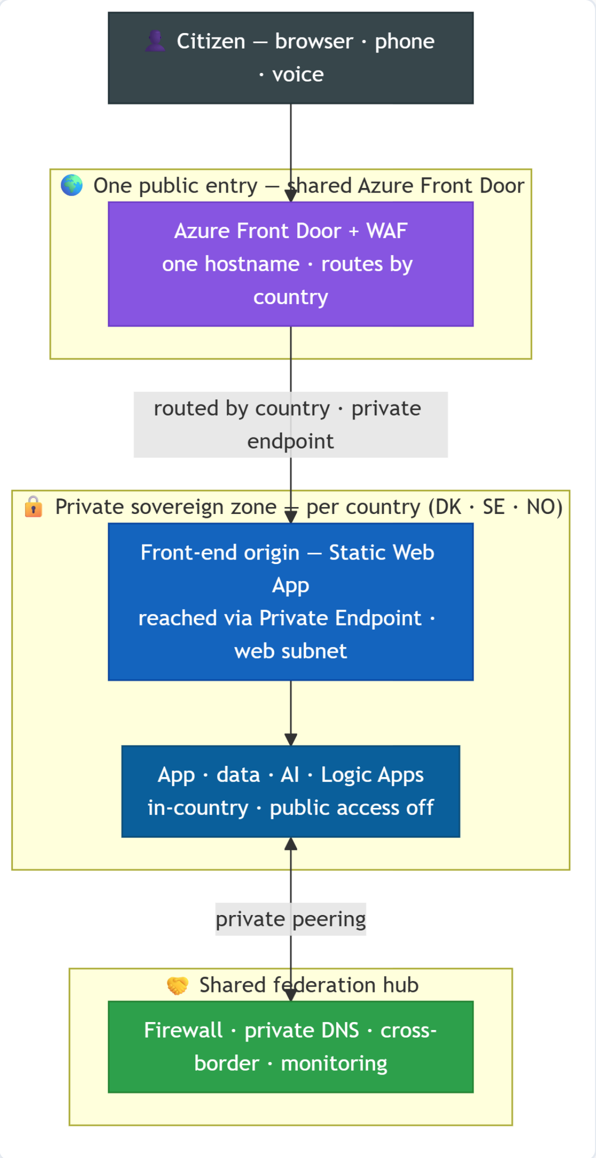


> Logic Apps are the **bridge** — stateful workflows that pre-fill the national form, submit it, and read the status back, writing each case into **Dynamics 365** and reaching authorities over **mutual TLS**. In production they're **VNet-integrated**.

How the network fits together — simply.

One entry, one experience — private and sovereign underneath.

- **One public entry**: a single **Azure Front Door** (web firewall) and one UI codebase — the citizen always sees the same portal.
- **Sovereign by design**: each country has its *own* private front-end origin — a **Static Web App reached through a private endpoint** in that country's `web` subnet, with no public origin.
- That `web` subnet in each spoke is the **private door** to the front-end, not a separate copy of the app.
- App, data, AI, Logic Apps are private and in-country too; a **shared hub** handles firewall, DNS, cross-border and monitoring.



Design patterns we use.

Agents-as-Tools
On a phone call, the speech model decides by itself when to call a function or pass to a human.

Gateway + Saga
One API gateway per country. The cross-border case is a 6-step workflow that rolls back if a step fails.

System of record — Dynamics 365
Dynamics 365 holds each case. The app writes through workflows and reads through the gateway with a cache.

Strangler fig
The caseworker app writes to a simple Dynamics 365 table today, the final case entity tomorrow — one switch.

Sidecar
The voice service pairs a call handler with a real-time audio bridge.

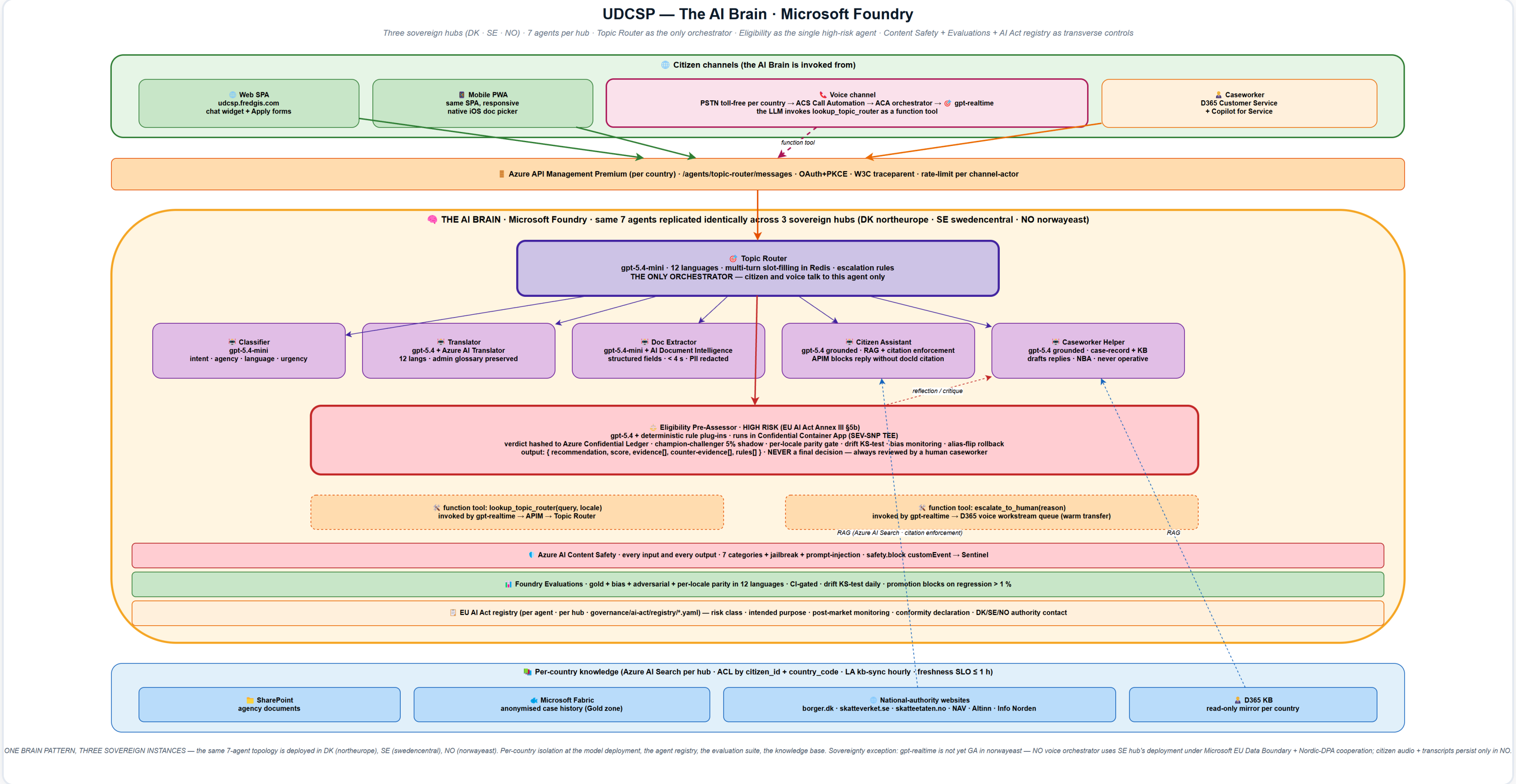
Circuit breaker
A failing national endpoint switches to a human queue — citizens never see a timeout.

The AI Brain.

One brain on Azure AI Foundry · one orchestrator · six specialists · safe by design

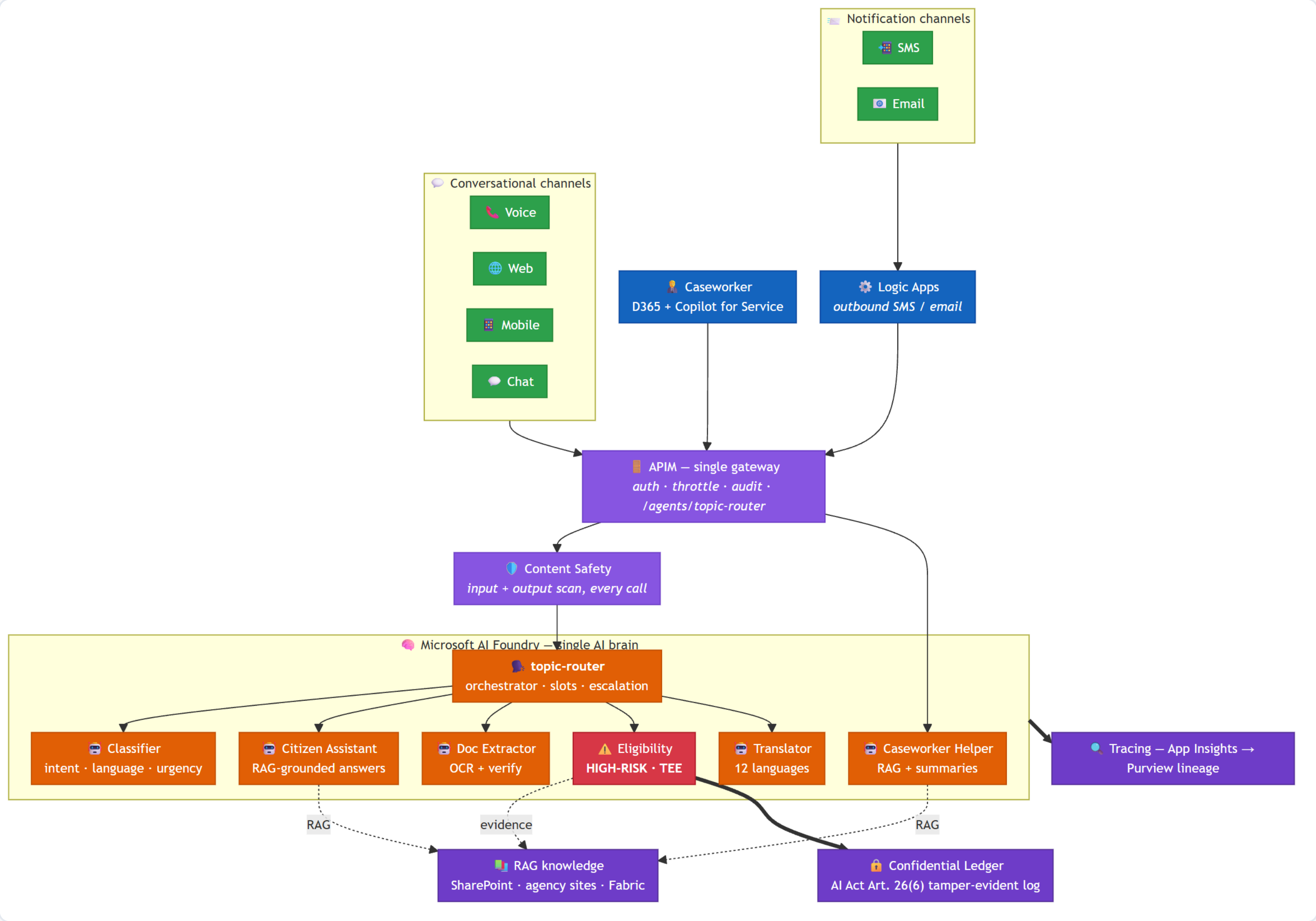
02

The AI Brain.



One brain, not seven chatbots. Every model call goes through Azure AI Foundry — so orchestration, safety, evaluation, tracing and the EU AI Act register live in one place. The **Topic Router** gives each task to the right specialist, and the whole brain is copied per country.

The AI Brain — every channel, one gateway.



How a citizen's question flows through the brain.

01 **The channel asks the brain**

Web, mobile and phone reach the Topic Router through the one private gateway — never a model directly.

02 **The router understands and remembers**

It finds the intent and the language across 12 languages, and keeps the short conversation for 24 hours.

03 **It passes the task to a specialist**

Classifier, Translator, Document Reader, Citizen Assistant, Caseworker Helper — or the high-risk Eligibility agent.

04 **The specialist does one job well**

The Assistant answers only from cited sources; the Reader extracts a payslip; Eligibility proposes a result with evidence.

05 **The answer returns — traced and sourced**

Every reply carries a trace number and its sources. On voice, the model decides when to call a tool or a human.

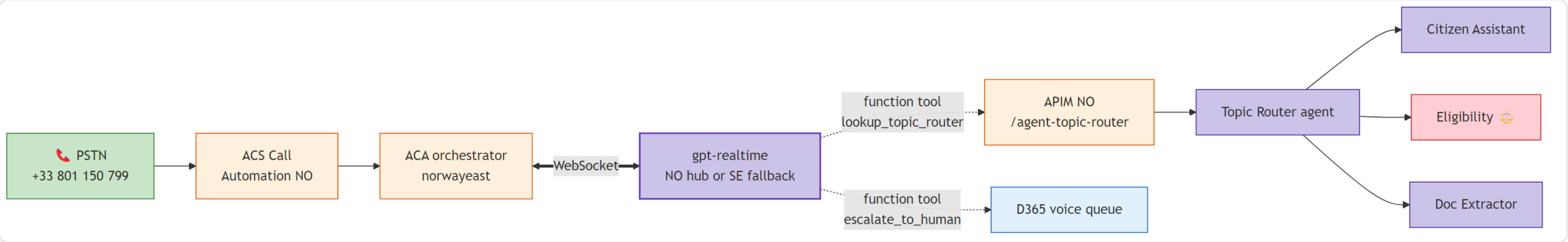
> Six specialist agents, two function tools, one orchestrator — seven agents in all, always under one human caseworker.

Agent catalogue.

Agent	Purpose	Model	EU AI Act class
Topic Router	Orchestration · 12 languages · slot-filling	Fast	Limited
Request Classifier	Intent · agency · language · urgency	Fast	Limited
Translator	12 languages · keeps admin terms exact	Strong + Translator	Limited
Eligibility Pre-Assessor ⚖️	Result + evidence · runs in a protected enclave · ledger-logged	Strong + rules	**High (public service)**
Citizen Assistant	Answers only from the cited public knowledge base	Strong, grounded	Limited
Document Reader	Passport / payslip / lease extraction	Fast + Doc Intelligence	Limited
Caseworker Helper	Summarise · draft replies · suggest next action	Strong, grounded	Limited

No agent makes a final decision. Every result is a *proposal* to a human caseworker, who confirms, changes or rejects it.

Voice channel — the model uses tools on its own.



> Citizen dials → the call lands on Azure Communication Services → a small service opens a two-way audio stream to the speech model. **The model decides on its own** to call `lookup_topic_router` or `escalate_to_human` (a warm transfer). This is the Microsoft Agent Framework *Agents-as-Tools* pattern, on a real phone call.

Eligibility model lifecycle.

01 **Champion-challenger**

A new version runs in shadow on 5% of traffic for one week, then we evaluate it.

02 **Parity gate per language**

A gold test runs in all 12 languages. If one language is too weak, the release is blocked.

03 **Drift detection**

A daily test on inputs and outputs. Drift opens a security incident and forces a new evaluation.

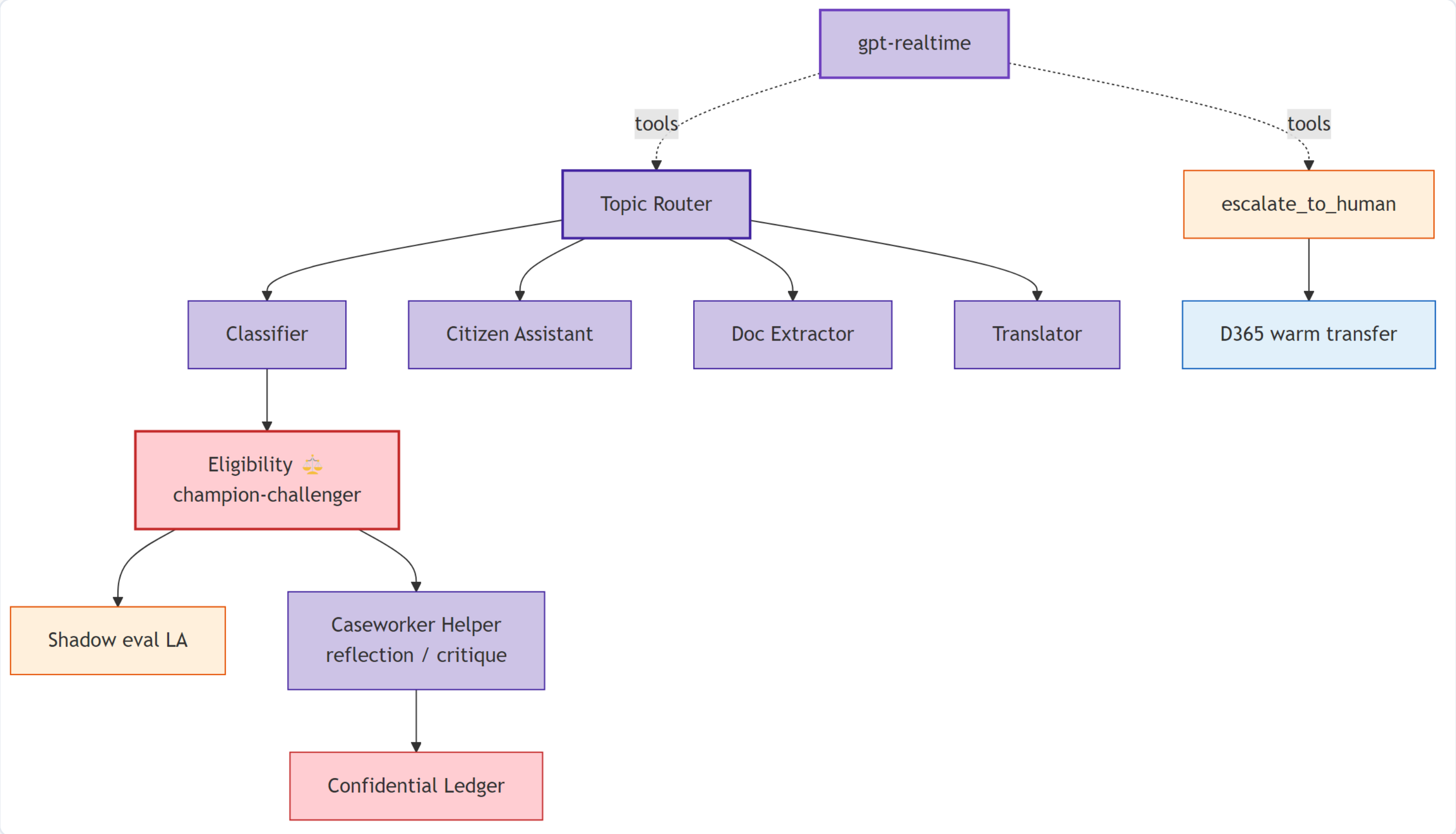
04 **Bias monitoring**

Fairness tests across age, location and channel for 30 days. A problem routes cases to a senior reviewer.

05 **Rollback in seconds**

Versions never change. Switching the alias restores the previous one at once, with an audit record.

Multi-agent coordination — more than a chatbot.



Handoff (router → six experts) · **Reflection** (the helper reviews the eligibility result) · **State graph** (the case is a 6-state workflow that can undo a step) · **Function tool** and **warm transfer** on voice · **Shadow / canary** for every model change.

Trust built into the brain.

Content Safety
A jailbreak and prompt-injection detector runs on every turn; a block raises a security event.

Evaluations
Quality is tested on every change — language parity, grounding and safety.

AI Act register
Each agent is declared with its purpose, risk level and limits, in a versioned file.

Confidential Ledger
Each high-risk result is hashed into a log that cannot be changed.

Human-in-the-loop
No agent decides. A caseworker confirms, changes or rejects every result.

Transparency
The citizen is always told AI is used — a chat badge and a spoken message in 12 languages.

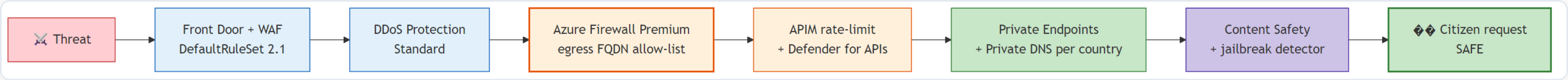
The Eligibility agent is the only high-risk part. It runs inside a **protected enclave** — its data is encrypted in memory, even from an administrator — and it never decides alone.

Security & compliance.

How we made it provable — eight EU laws mapped to controls in code

03

Defence in depth — many layers.



Edge — application

Azure Front Door and web firewall — common-attack rules, bot protection, rate limits

Edge — network

Standard DDoS protection on every network

Traffic out

Azure Firewall · destination allow-lists · TLS inspection

Identity & data sovereignty.

IDENTITY

- One citizen-identity system per country (Microsoft Entra External ID)
- National eID via a certified broker — MitID · BankID + Freja+ · ID-porten
- Support for the EU Digital Identity Wallet
- Admins get time-limited access only, through one jump-host

DATA — FIVE ZONES, ALL IN-COUNTRY

- **Live** — managed database and cache
- **Documents** — write-once, cannot be changed
- **Conversations** — kept for AI Act records
- **Knowledge** — search, locked per citizen and country
- **Analytics** — sovereign EU Fabric capacity

> One encryption key set per country · modern TLS in transit · two-way TLS to partners · field-level encryption for national IDs · public network access **turned off** everywhere.

How we made it compliant — the story.

01 We started from the law, not the features

Eight bodies of EU law touch one interaction — a single voice call is data protection, accessibility, e-identity, cybersecurity and AI law, all at once.

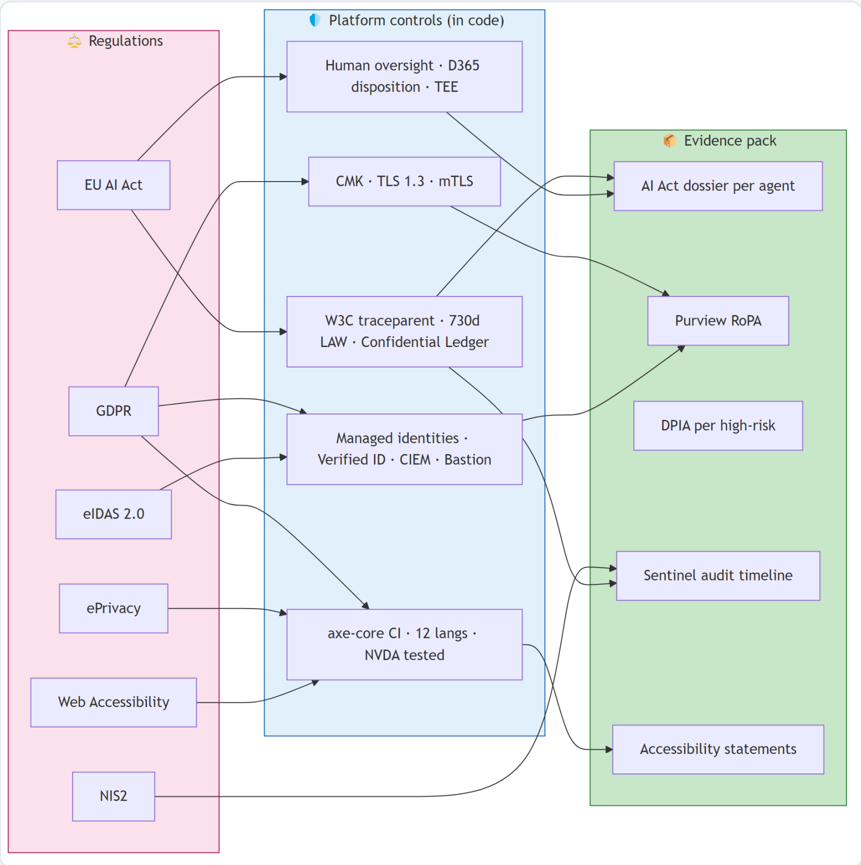
02 Each obligation became a control in code

Every rule maps to a real platform mechanism, with an evidence file behind it.

03 We made the evidence provable

One trace number follows every interaction, from the browser to the model and back, kept for two years.

GDPR — data protection · EU AI Act — high-risk AI · eIDAS — e-identity · NIS2 — cybersecurity · ePrivacy — cookies · WCAG — accessibility



EU AI Act — article by article.

Article	Requirement	UDCSP delivery
Art. 12	Automatic records ≥ 6 months for high-risk	Model logs kept **730 days** = 2× the minimum
Art. 13	Transparency for deployers	A register entry per agent · tests per language
Art. 14	Human oversight	A caseworker confirms every result · ledger-logged
Annex III §5(b)	High-risk = access to essential public services	The Eligibility agent is declared high-risk
Art. 50	The citizen must know AI is used	AI badge on chat · spoken message in 12 languages

> The AI Act, GDPR, ePrivacy, eIDAS, NIS2 and accessibility law are each mapped to platform controls. The full mapping is in `docs/biz/datacompliance.md`.

Security in action — a prompt injection, stopped.

A message that tries to leak the hidden instructions, or change a result, is caught at **three independent layers** — three different Azure tools:

01 **Azure API Management**

The gateway sees the strange token rate and blocks the data leak.

02 **Azure AI Content Safety**

The jailbreak detector raises a security event and opens an incident.

03 **Fixed eligibility rule**

A deterministic plug-in rejects the request before any model runs.

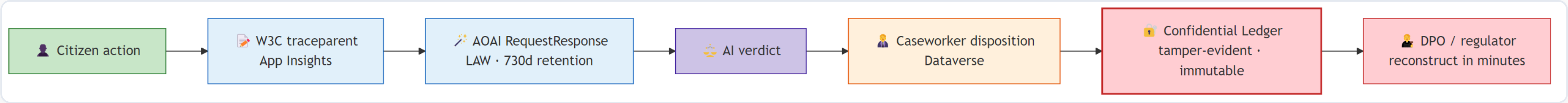
THE AUTOMATIC RESPONSE

- The session is isolated at once
- The citizen flow is brought back safely
- An audit pack is saved for review

0

citizen data exposed

Compliance in action — replay a decision, months later.



Hans, a data-protection officer, takes the citizen's **trace number** and rebuilds the whole decision — the model, the tokens, the result, the human choice, and the cryptographic ledger proof.

The AI Act minimum is six months. We keep **two years**. From the request to a full audit pack: **under 10 minutes**.

Monitoring & FinOps.

One anonymized central view · end-to-end tracing · cost per token

04

One central view — anonymized, in Fabric & Power BI.

Leadership needs the big picture. But raw citizen data never leaves its country. So only **anonymized, aggregated numbers** travel to a central view.

12,480
cases this month

4 days
median time (was 28)

71%
AI-assisted, human-approved

4.5 / 5
citizen satisfaction

CASES BY COUNTRY (ANONYMIZED)

DK DK
SE SE
NO NO

BY SERVICE

Residency 46%
Tax 28%
ID & docs 26%

MEDIAN DAYS TO DECISION

Q1 · 28
Q2 · 15
Q3 · 8
Now · 4

Only anonymized, aggregated numbers leave each country — never a name, never a case file. Built on Microsoft Fabric and Power BI · refreshed every hour.

FinOps — cost under control.

ALLOCATION & SHOWBACK

- Every resource tagged `country`, `workload`, `cost-center`
- Cost views per country and per workload
- Each ministry sees its own country's cost

TOKEN BUDGET & CAPACITY

- A monthly token budget per agent, written in code
- The build **fails** if the total is over the pool
- Reserved capacity for steady models · pay-as-you-go for peaks
- Alert on a +30% jump in one day

> We treat AI tokens like any other budget line: planned, tagged, and checked at build time.

Performance & reliability.

Fast for the citizen, always available — measured targets, automatic failover, chaos drills

05

Performance & reliability.

99.9%
citizen web target, per country

≤ 2 s
voice answer, 95% of calls

15 min
recovery point — max data loss

4 h
recovery time to the backup EU region

Active-passive per country

Azure Front Door sends traffic to the healthy region. Failover in under 5 minutes.

Backup & disaster recovery

Database, cache, storage and agent hosts. One recovery vault per country.

Chaos drills

We break things on purpose — region failover, network cut, database failover. Monthly in test, quarterly in production.

Autoscale & caching

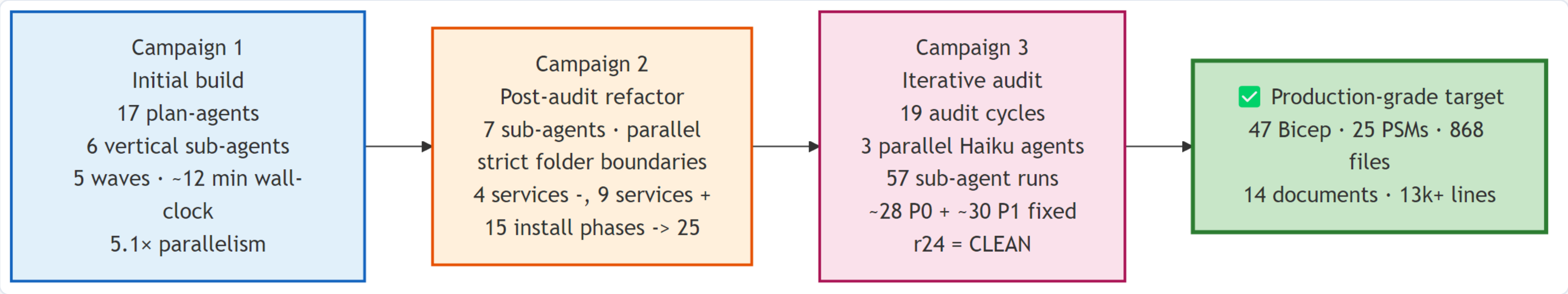
Services scale out under load. Reads are cached at the gateway, so peaks stay fast.

How we built it.

The platform itself was built by an agent swarm — in under 45 minutes of wall-clock time

06

UDCSP was built by an agent swarm.



3
multi-agent campaigns

~5×
faster than sequential

24
audit rounds to first clean

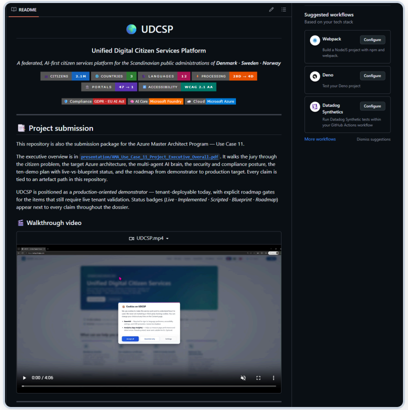
868
tracked files produced

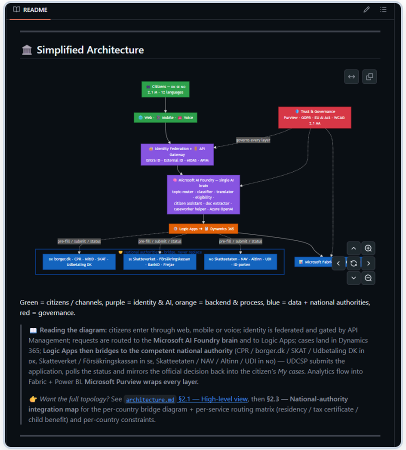
> Three campaigns — build, refactor, then repeated audits — produced the platform. Six sub-agents owned separate folders and ran in parallel. Each round produced fix commits until round 24 was the first fully **clean** one. The one-command installer is the executable summary of it all.

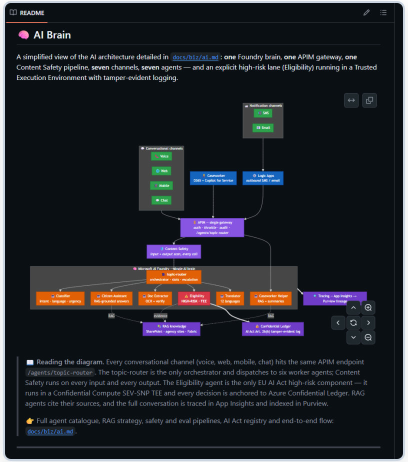
Everything is in the open — one repository.

THE STORY STARTS IN THE DOCS

- `/docs/biz` — the business story: who we build for, every channel, data-and-compliance, demo scenarios, and an acceptance recipe for "done".
- `/docs/tech` — the engineering: architecture deep-dive, data model, network design, one-command install guide, disaster-recovery runbook.
- `/infra` — the Bicep that builds every zone · `/scripts` — the installer.







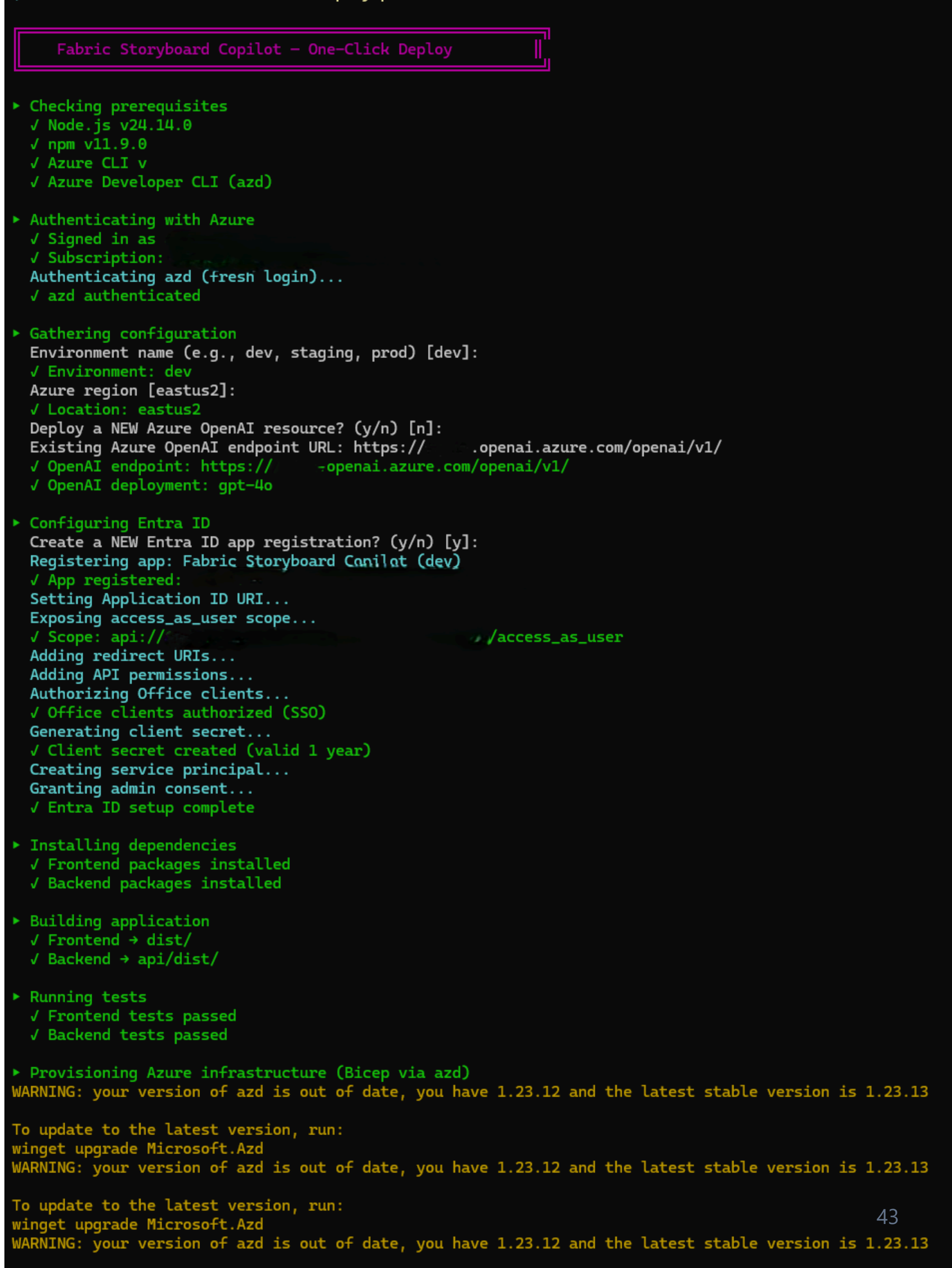
SHORT LINK
aka.ms/UDCSP

GITHUB
github.com/fredgis/UDCSP

One idempotent deploy — provision.

One script, one click — and you can run it again safely.

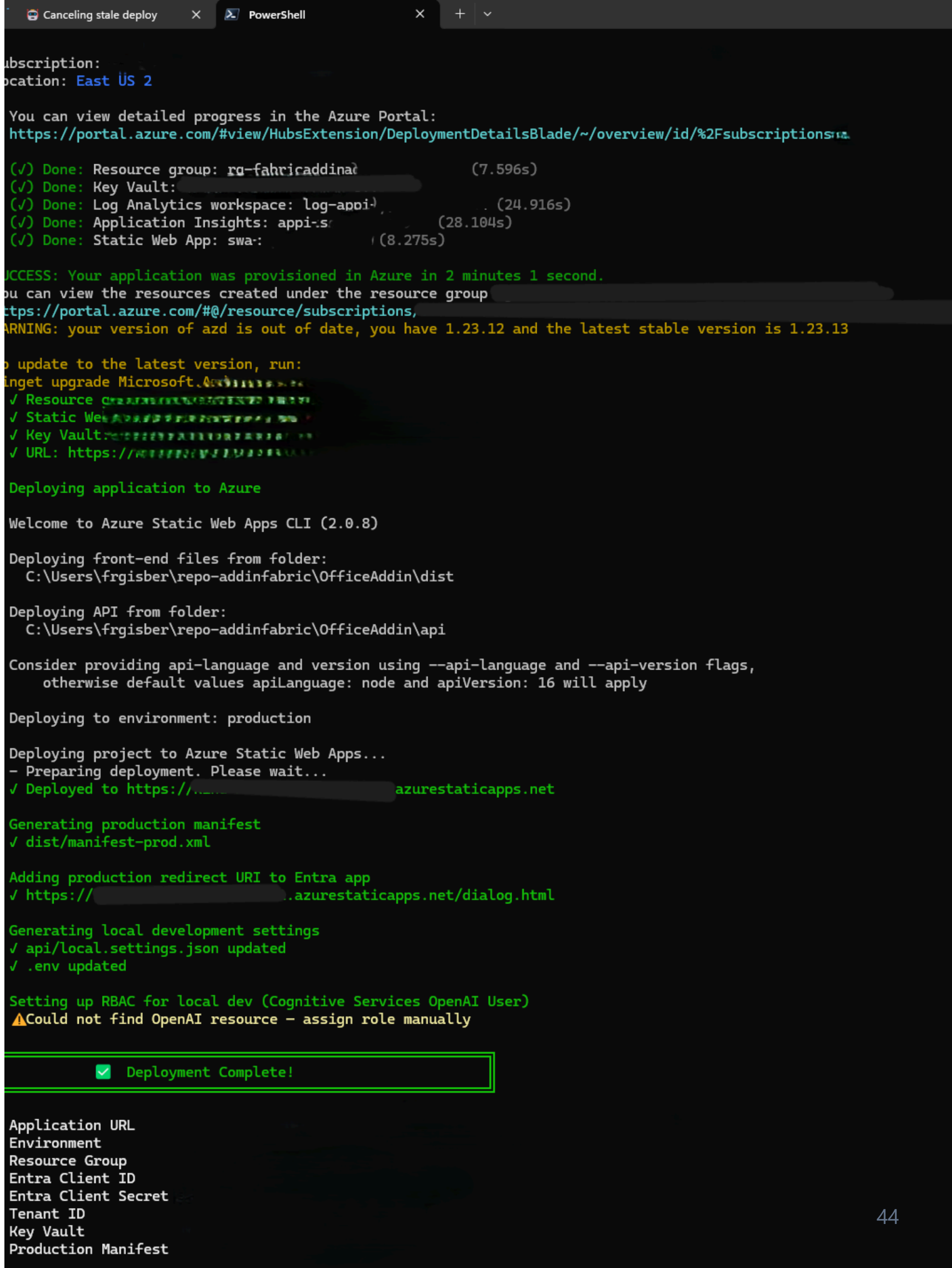
- Checks prerequisites, signs in to Azure, registers the **Entra ID** identity app.
- Installs, **builds**, and runs the **tests** before anything is deployed.
- Provisions all the infrastructure **as code** — Bicep, through `azd`.
- **Idempotent**: a second run reuses what exists — never a duplicate.



One idempotent deploy — release and verify.

It deploys, verifies, and reports — every time the same way.

- Provisioned in **~2 minutes**, then the app is deployed to Azure.
- Generates the **production manifest**, wires redirect URIs and **RBAC** (role-based access control).
- Ends with **Deployment Complete** and a summary of every resource and URL.
- The same script runs in the **pipeline on every change** — repeatable, not hand-made.



Real scenario.

Life imitates the demo — a story we did not plan

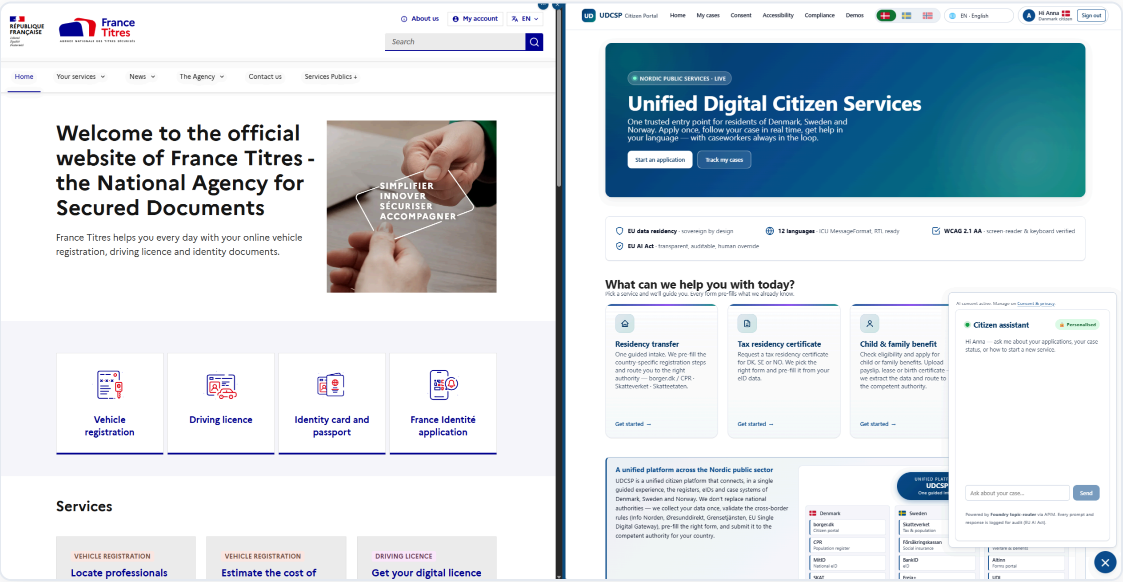
07

"I bought my car back." 🚗

To change my licence plate, I signed in to the **official French government portal** — France Titres.

I had **never seen it** while building UDCSP.

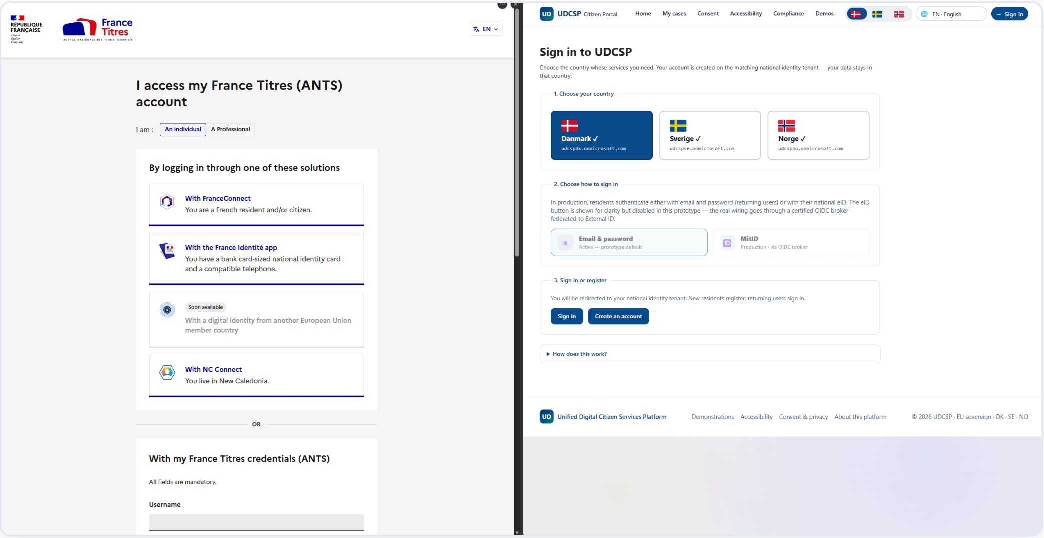
And there it was — the same one-front-door layout, the same service cards (vehicle registration, driving licence, ID & passport), the same assistant in the corner. 😄

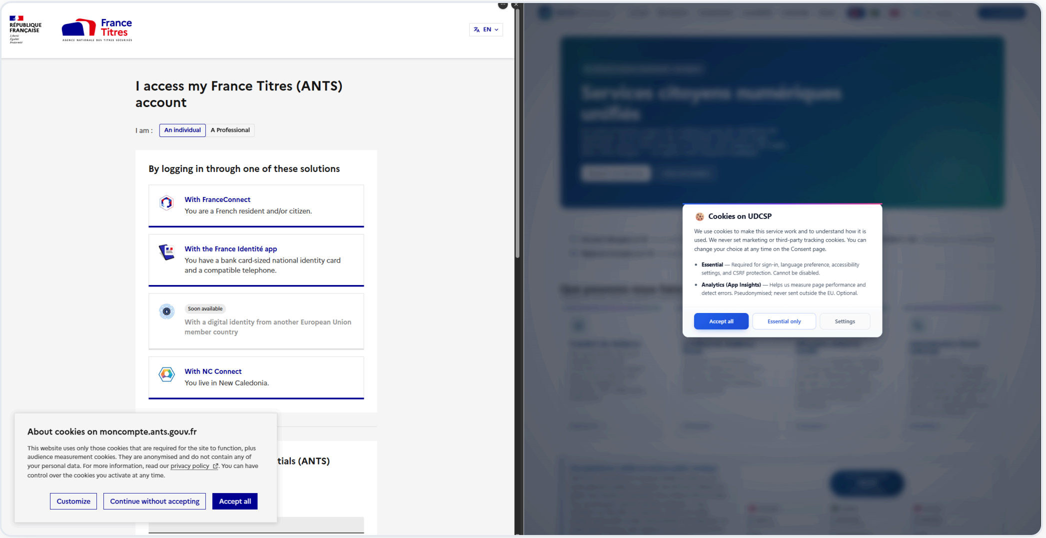


FR FRANCE TITRES — THE REAL THING

UDCSP — OUR DESIGN

…and we'd built it first. 🙂





Pick your country, sign in with a national eID, and — "*coming soon*" — an **EU cross-border identity**. That is our model, almost line for line. **We designed it before we ever saw it.** 🙂

Status & roadmap.

What runs today · what's blueprint · how it reaches production

08

Requirement coverage — the honest scorecard.

Unified identity
Entra External ID · eIDAS-ready
Live · Verified ID

12-language portal
WCAG 2.1 AA accessibility
Implemented

AI assistant
web · mobile · voice
Live

Cross-border case
DK → SE pre-fill + saga
Live · saga

Eligibility + human-in-loop
verdict, then a person decides
Implemented · live TEE

EU AI Act evidence
article-by-article trail
Implemented · live ledger

GDPR rights
records · access · erasure
Scripted

Sovereign monitoring
9 workbooks, per country
Live · Fabric F64

One-command deploy
25 phases, idempotent
Live

Live — today · Implemented — merged & tested · Scripted — idempotent installer · Blueprint — designed & gated

The working core carries the demo; each blueprint brick has a named gate.

Four gates from accelerator to production.

GATE 1 · WK 1–2

Tenant validation

Confirm model deployments, Fabric F64 capacity, DDoS plan & Foundry hub.

GATE 2 · WK 3–6

Live confidential compute

Eligibility on a SEV-SNP enclave + Confidential Ledger. **Demo 6 → Live.**

GATE 3 · WK 6–14

Partner-agency integration

Mutual-TLS agreements + production federation hub. **Demo 1 → real authority.**

GATE 4 · WK 14–20

D365 + Copilot for Service

Strangler-fig repoint of case writes. **Demo 5 → Live.**

Today → Gate 1 → Gate 2 → Gate 3 → Gate 4 → production. **Each gate is independently shippable** — every blueprint brick has an owner, a window, and a demo it flips to live.

Demo.

10 minutes, live on a real tenant

09

The run of show.

01 **Anna moves DK → SE** Live

Sign in with a Danish eID · upload a passport and lease · the AI reads, translates and proposes a result · consent · the case crosses the border.

02 **Lars calls the voice line** ⭐ Live

A blind citizen dials a real free number · the model answers in Norwegian and routes itself · a human transfer is offered.

03 **Maria uses a screen reader in Polish** Implemented

The whole portal in Polish, keyboard only, screen-reader clean — accessibility as a citizen right.

04 **Erik photographs a payslip on iPhone** Live

The same web app on mobile · native iOS picker · structured fields and a result, inline.

05 **Ole builds it from a clean tenant** Live

One command: 25 phases, sample data seeded, smoke tests green.

> Hero moment — **a real phone call**: dial `+33 801 150 799`, hear the model answer, and watch the live dashboard update within two minutes.

One command, from clean tenant to running platform.

```
git clone https://github.com/fredgis/UDCSP
cd UDCSP
pwsh ./scripts/install/Install-UDCSP.ps1 `
     -Environment dev -Zone all `
     -SeedSyntheticData -Verbose
```

- 25 phases in order: landing zone → identity → security → data → monitoring → gateway → Foundry → Dynamics → frontend → voice → governance → quality
- Sample data is seeded in parallel with the frontend
- A smoke test runs at the end (identity · gateway · AI · case creation · dashboard · accessibility)
- An HTML report is saved in `scripts/install/reports/<timestamp>/`

> From `git clone` to a working federated platform with real sample data: **one command**. The same script runs in our pipeline on every pull request.

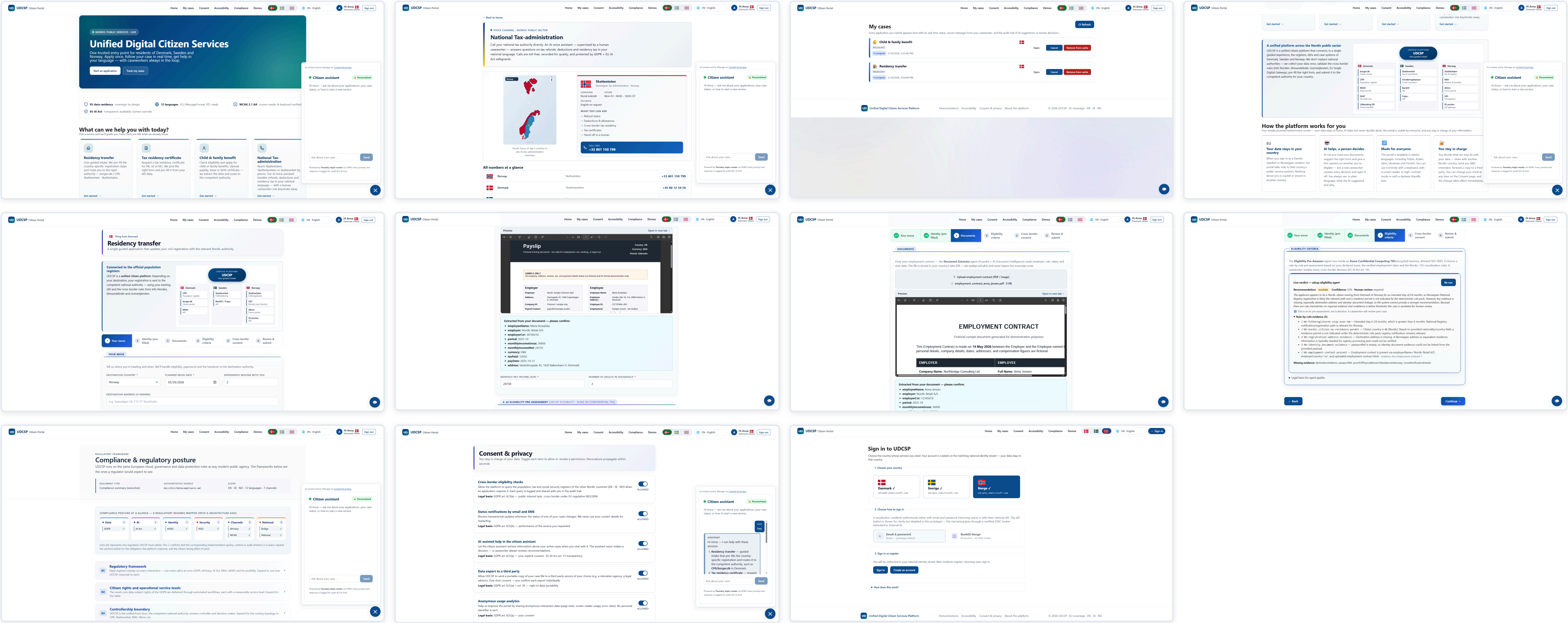
CLOSING

UDCSP — a production accelerator with a named path to production.

3 sovereign zones · 7 AI agents · one private path from the portal to the case · every decision anchored to a regulation · every claim labelled and provable on a real Azure tenant. A working core you deploy with one command today — and a four-gate roadmap that takes the blueprint bricks to production. A real French government portal, seen by accident, proved the architecture was right.

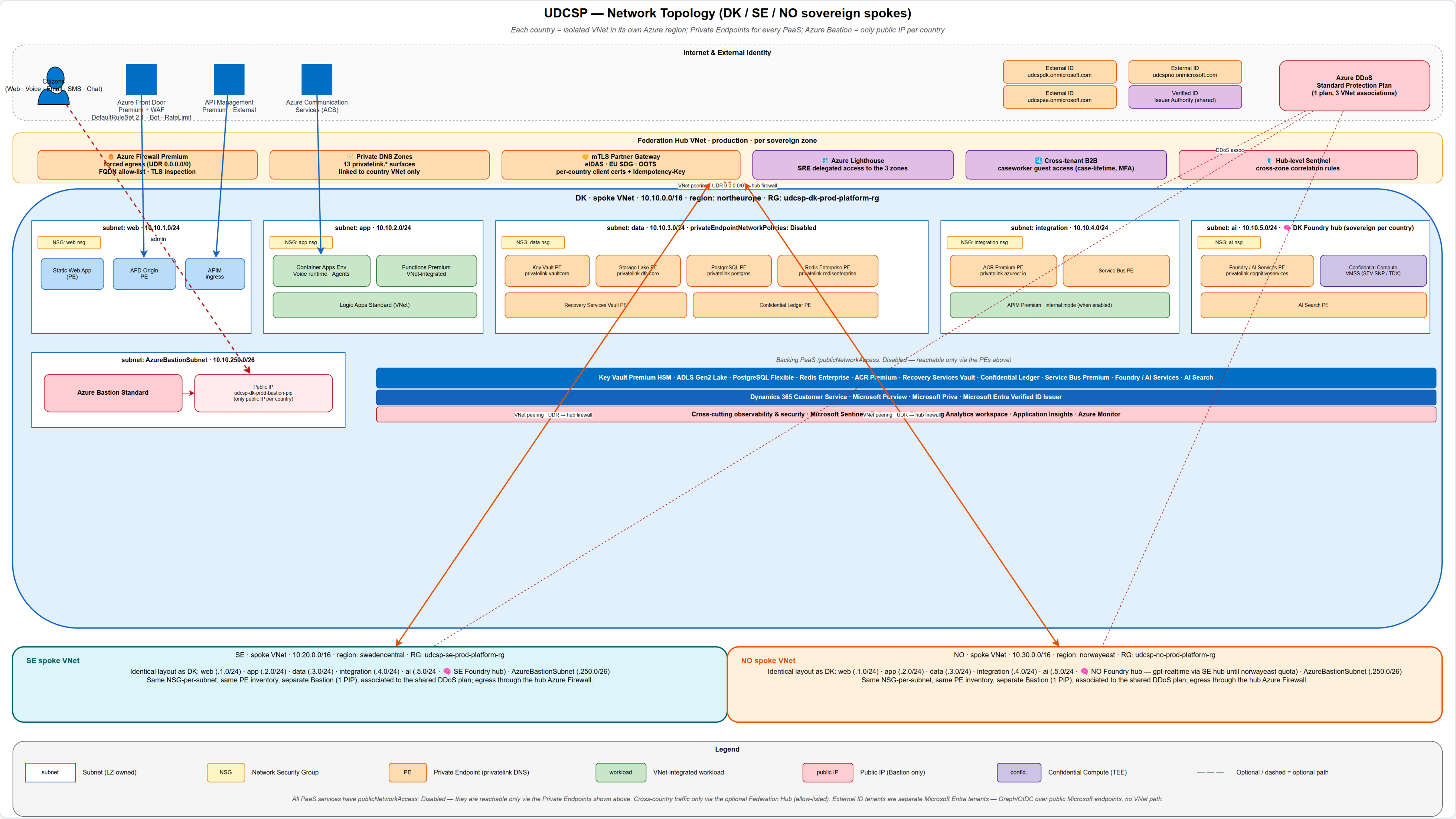
Frederic Gisbert · June 2026 · github.com/fredgis/UDCSP

Annex — the platform, screen by screen.



1 Home · **2** Contact / voice · **3** My cases · **4** Assistant · **5** Residency intake · **6–7** Document reading · **8** Cross-border result · **9** Compliance · **10** Consent · **11** Sign-in. One web app, twelve languages, three channels — the same React and TypeScript code on desktop, tablet and phone.

Annex — network topology in full.



Annex — telemetry pillars in full.

METRICS / LOGS

Azure Monitor

3 workspaces, one per country · 180 days hot · 7 years cold · never joined

DISTRIBUTED TRACE

Application Insights

One trace number end-to-end · front door → gateway → workflow → Dynamics → agent → model

AI QUALITY

Foundry Evaluations

Continuous · drift monitors · language parity · bias monitoring

AI ACT EVIDENCE

Confidential Ledger

Records that cannot be changed · joins logs, AI and Dynamics in one query

ACTIVE

Synthetic + real-user

5 external regions · 60-second probes · load times per language

DASHBOARDS

9 workbooks

platform health · citizen-journey funnel · AI-decision traces · per country